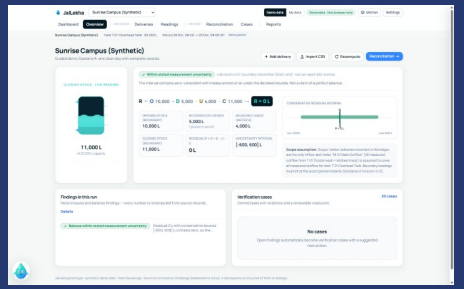
JalLekha

A water-balance engine that reconciles tanker receipts, tank readings and meter logs - what came in, what was measured, what needs checking.

Sector: Sustainability - Energy, Climate and Others
Sub-theme: pollution mitigation, water conservation, waste management
Track A - Product / Technology Solution · Open Level

Working prototype - live demo

- **Deterministic engine - AI never decides.**
- **55 unit + 33 e2e checks, all green.**
- **Browser demo · synthetic data.**



jallekha.vercel.app/demo

2

The Problem

What real challenge in society, government service or industry are you solving?

What is the problem?

Tanker receipts, tank dipstick readings and cumulative meter logs live in three separate registers. When water doesn't add up, staff trace entries by hand - and the register cannot say why.

Who is affected, and where?

Campus facilities teams and housing-society maintenance staff; students and residents feel it in billing disputes and unreliable supply. Target: urban Maharashtra.

Evidence

Honest by design: no field pilot yet. The engine is validated on synthetic ledgers - 55 unit tests and 33 browser checks assert the four signature scenarios to the litre (docs/VALIDATION.md).

Current gaps

Registers total numbers but flag nothing: duplicates go unnoticed, missing readings read as losses, and corrections overwrite the audit trail - disputes cannot be evidenced.

3

separate registers - one water balance that must agree

Why it matters

Unmeasured losses stay invisible; supplier disputes stay unresolvable.

3 Our Solution & Innovation

What have you built or designed, how does it work, and what is new about it?

Our solution in brief

Facilities teams record tanker receipts, tank readings and meter values; a deterministic engine reconciles them into one water balance with a visible uncertainty interval - and findings become owned, evidence-backed cases.

How it works

1 Input

Receipts, dipstick readings, meter logs - typed or CSV

›

2 Process

Engine computes R = O + D - U - C with summed bounds

›

3 Output

Verdict, findings, owned cases, audit trail

What makes it new?

	Existing approach	Our approach
Cost	*₹ 0 software + hours of tracing*	*₹ 0 licence - any browser*
Speed / time	*Days of hand-tracing per dispute*	*Seconds to a documented verdict*
Reach / access	*One register-holder's memory*	*Any operator; CSV in, review out*
Ease of use	*Spreadsheet discipline required*	*Guided demo; plain-language verdicts*

Key USP

Uncertainty you can see and verdicts that never accuse - same inputs, same verdict, and every correction keeps its before-and-after history.

4 Feasibility & Implementation

How ready is the solution, and what will it take to implement it on the ground?

Current stage (mark ✔ where you are today)

- Idea
- Proof of Concept
- **Working Prototype** ●
- Field Pilot
- Ready to Deploy

Technology

Next.js 16 + TypeScript web app; pure-TypeScript rule engine recon-v1.0; CSV import; browser-local storage; optional provider-neutral AI, off by default.

Estimated cost

₹ 0 licence - runs in a browser on one existing device. Pilot needs operator time only: no sensors, no hardware.

Timeline

Now: 4-6 week single-device pilot. Next: shared authenticated workspaces (adapter designed). Then: multi-site rollup.

Partners

Seeking one permissioned campus or society; measurement-quality mentorship; a route toward a government-department pilot.

5

Impact & Scalability

What change will your solution create, and how can it grow beyond the first location?

1

Beneficiaries reached

pilot campus or society in year one

4

Saving in cost / time

impact metrics instrumented - baselines set on field data

3

Jobs or livelihoods

sustainability domains evidenced: water, haulage, trust

Social impact

Evidence-backed records end billing disputes and repeat reconciliation work; residents see a documented balance instead of a disputed bill.

Sustainability

Zero-licence software on existing devices. The pilot publishes baselines first - no litres-saved claims before field data exists.

Scale-up plan

NOW: single-device pilot > NEXT: shared authenticated workspaces (adapter designed) > THEN: multi-site rollup for a campus chain or ward.

6 Team/Individual & Support Required

Who is behind the idea, and what help do you need from the Challenge?

AJ

Arya Jannawar

Team Lead

Product & the reconciliation equation

SP

Siddhant Pardeshi

Quality & user workflow

55-test gate, 33 browser journeys

AV

Avani Jannawar

Engineering lead

Engine, CSV pipeline, adapters

VS

Vijay Sahu

Research & validation

Measurement uncertainty & evidence

Mentor: to be assigned - seeking measurement-quality and study-design guidance.

Thank you | Team SevaForge · github.com/Avani376/jallekha · jallekha.vercel.app

Support we are seeking

- [ ] One permissioned pilot campus or society
- [ ] Mentorship: study design & measurement quality
- [ ] Lab access for reviewer-time measurement
- [ ] A route to shared, authenticated deployment
- [ ] Introduction to a department for a broader pilot

A 4-6 week pilot on one device publishes the first honest baselines - turning the prototype into evidence.